

Bachelor of Technology — Data Science and Engineering

Table 1: Data Science and Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	33
	Discipline Electives	33
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Data Science and Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-213	Reverse Engineering	0-0-2-1	1
2	DS-201	Data Handling and Vizualisation	2-0-2-3	3
3	DS-301	Mathematical Foundations of Data Science	3-1-0-4	4
4	DS-302	Computing Systems for Data Processing	2-0-2-3	3
5	DS-313	Statistical Foundations of Data Science	3-1-0-4	4
6	DS-404	Information Security and Privacy	3-0-0-3	3
7	DS-412	Matrix Computations for Data Science	3-0-2-4	4
8	CS-305	Artificial Intelligence	3-0-0-3	3
9	DS-413	Introduction to Statistical Learning	3-1-0-4	4
10	DS-411	Optimization for Data Science	3-1-0-4	4
Total				33

Table 3: B. Tech Data Science and Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		HS-342	German I	3-0-0-3	
10		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Data Science and Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
9	IKS + HSS (choose 0–3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0–3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22–25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Data Science and Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	CS-213	Reverse Engineering	0-0-2-1	1
3	DC	DS-201	Data Handling and Vizualisation	2-0-2-3	3
4	DC	DS-301	Mathematical Foundations of Data Science	3-1-0-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Data Science and Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	DS-302	Computing Systems for Data Processing	2-0-2-3	3
3	DC	DS-313	Statistical Foundations of Data Science	3-1-0-4	4
4	DC	DS-404	Information Security and Privacy	3-0-0-3	3
5	DC	DS-412	Matrix Computations for Data Science	3-0-2-4	4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Data Science and Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-305	Artificial Intelligence	3-0-0-3	3
2	DC	DS-413	Introduction to Statistical Learning	3-1-0-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Data Science and Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-411	Optimization for Data Science	3-1-0-4	4
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					8

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Data Science and Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	DS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Data Science and Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	DS-010	Internship		2
2	MTP	DS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 33 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Data Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-507	Probabilistic Robotics	3
2	AR-509	Deep Learning for Robotics	4
3	AR-516	Introduction to Blockchain and Web3	3
4	AR-517	Introduction to Cyber security	3
5	BE-303	Applied Biostatistics	4
6	BE-304	Bioinformatics	4
7	BE-502	Design and Analysis of Bioalgorithms	3
8	BE-505	Computational Biology	3
9	CS-212	Design and Analysis of Algorithms	4
10	CS-214	Computer Organization	4

Data Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	CS-241	Introduction to Cryptography	3
12	CS-302	Paradigms of Programming	4
13	CS-303	Software Engineering	3
14	CS-304	Formal Languages and Automata Theory	3
15	CS-306	Operating Systems	3
16	CS-307	Systems Practicum	2
17	CS-308	Large Application Practicum	2
18	CS-308P	Large Application Practicum	2
19	CS-309	Information and Database Systems	4
20	CS-310	Introduction to Communication and Distributed Processes	4
21	CS-313	Computer Networks	4
22	CS-347	Software Engineering	4
23	CS-350	Computer Graphics	3
24	CS-401	Architecture of High Performance Computers	3
25	CS-402	Compilers	2
26	CS-403	Algorithm Design and Analysis I	3
27	CS-405	Verification of Reactive Systems	3
28	CS-406	Computer Networks	4
29	CS-451	Computer Graphics and Game Design	3
30	CS-456	Distributed Databases	3
31	CS-501	Access Networks	3
32	CS-502	Compiler Design	4
33	CS-506	Cognitive Modeling	3
34	CS-507	Computer Architecture	4
35	CS-508	Introduction to Network Security	3
36	CS-510	Randomized Algorithms	4
37	CS-511	Applied Probability	2
38	CS-512	Matrix Theory	2
39	CS-513	Discrete Mathematics	4
40	CS-514	Advance Data Structure and Algorithm (CS-580)	4
41	CS-515	Advanced Computer Science Practicum	3
42	CS-516	Exploratory Project	3
43	CS-520	Introduction to Quantum Computing	3
44	CS-521	Introduction to Post Quantum Security	3
45	CS-522	Distributed Algorithms	3
46	CS-524	Computational Complexity Theory	4
47	CS-525	Computational Foundations of Cryptography	3
48	CS-526	Mathematical Foundation of Cryptography	3
49	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3

Data Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	CS-541	IoT systems and the Cloud	3
51	CS-542	Design Patterns for Scalable Systems	3
52	CS-544	Formal Concept Analysis: Theory and Practice	3
53	CS-545	Software Design Patterns	3
54	CS-546	Design of Concurrent Software	3
55	CS-547	Network Management Systems	3
56	CS-548	Cloud Networking	3
57	CS-549	Performance Analysis of Computer Networks	3
58	CS-550	Computer Graphics and Geometric Design	3
59	CS-551	Human Computer Interaction	3
60	CS-560	Text Retrieval and Mining	3
61	CS-561	Map Reduce and Big Data	3
62	CS-563	Scalable Data Science	4
63	CS-570	Algorithms for Curves and Surface	3
64	CS-571	Programming Practicum	3
65	CS-591	Special Topics in Knowledge Discovery and Data	1
66	CS-592	Selected Topics in Artificial Intelligence	2
67	CS-601	Probability and Random Processes	4
68	CS-603	Managerial Decision Making	3
69	CS-606	Cognitive Modeling	3
70	CS-609	Speech processing	3
71	CS-611	Program Analysis	4
72	CS-630	Speech Technology	4
73	CS-660	Data Mining for Decision Making	4
74	CS-661	Knowledge Representation and Reasoning	3
75	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
76	CS-670	Kernal Methods for Pattern Recognition	4
77	CS-671	Deep Learning and Applications	4
78	CS-672	Advanced Topics in Deep Learning	4
79	CS-673	Advanced Computer Vision	4
80	CS-677	Soft Computing	3
81	CS-683	Generative AI	4
82	CS-685	Natural Language Processing	3
83	CS-686	Large Language Models: Core Concepts to Custom	4
84	CS-693	Compressed Sensing and its applications	3
85	EE-203	Network Theory	3
86	EE-304	Communication Theory	4
87	EE-305	Digital Signal Processing	4
88	EE-511	Computer Vision	4
89	EE-522	Matrix Theory for Engineers	3

Data Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
90	EE-530	Optimization theory	3
91	EE-531	Estimation and Detection Theory	3
92	EE-534	Probability and Random Processes	3
93	EE-541	Tensors: Techniques, Algorithms, Applications for Signal	4
94	EE-543	Vision and Learning Based Control	3
95	EE-574	Biomedical Signal and Image Analysis	3
96	EE-575	Applied Statistics for Data and Signals	4
97	EE-608	Digital Image Processing	4
98	EE-620	Advanced Digital Signal Processing	3
99	EP-301	Engineering Mathematics - 2	4
100	MA-530	Graph Theory	4
101	MA-605	Statistical Data Analysis	3
102	ME-522	High-Performance Scientific Computing	3
103	ME-526	Programming Paradigm for Open-Source Software	1
104	PH-302	Introduction to Statistical Mechanics	3
105	PH-522	Statistical Mechanics	4
<i>Compulsory project components — counted as DE, not selectable</i>			
106	DS-498P	Major Technical Project - I (compulsory; counted as DE)	4
107	DS-499P	Major Technical Project - II (compulsory; counted as DE)	4